

Domain shift in distilled T5-based models for language translation tasks

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Abstract

This work investigates about black-box knowledge distillation from a GPT-5 Mini teacher for language translation tasks, by training four student models from the T5-Efficient and mT5 architectures on two different domains. Using only the predictions generated by the teacher, we evaluate in-domain and out-of-domain predictions, using chrF++ and BLEU metrics. Results show that models with more parameters have stronger translation and generalization quality, multilingual pretraining provides a significant advantage, and with mixed domain training you can mitigate domain shift by paying a small cost in overall performance.

Keywords: distillation, domain shift, language translation, t5, mt5

1 Introduction

In recent years, more and more cumbersome LLMs have come to life in the AI race led by OpenAI and joined by all other competitors and big techs like Google, Anthropic, and Meta, just to say some. These companies have created models with billions if not trillions of parameters [1], and are general purpose, which means they are capable of doing a wide range of tasks such as natural language understanding and generation, decision making, and reasoning [2].

The main downside of these models is also their main advantage: the size. Running inference requires expensive hardware in order to load all parameters into dedicated GPU memory. Moreover, if we want to perform a task that only uses a subset of the

model capabilities, all the parameters are still active, leading to a waste of memory and computational power.

In the case we only want to perform a certain task, it might be useful to condense a larger model into a smaller one, specialized on that specific task while preserving comparable performance. This process is commonly known as distillation [3], and consists in transferring knowledge from a teacher model into a smaller student model.

A limitation arises when the teacher model we want to use is proprietary, and internal parameters or output logits are not accessible. Using a black-box makes the distillation process more difficult, and in those scenarios alternative strategies must be considered. One of them is training the student using only teacher’s predictions; in this way we are still able to train the smaller model, but a reduction in performance is expected compared to full logits distillation [4].

In this paper, we investigate the distillation of task-specific student models from a GPT-5 Mini teacher using only its output predictions. We test two different student architectures across multiple parameter counts in order to assess their performance under domain shift. The models are trained on English-to-Italian translation and evaluated using BLEU and chrF++ metrics to measure both in-domain and out-of-domain effectiveness. Many more metrics like COMET and TER exist, but in order to keep things simple I decided to focus only on those two.

2 Methodology

2.1 Task

First, let’s define the task of interest. I wanted to train the model on something useful and challenging, in order to stress the student and understand its limitations. I opted for a translation task, specifically English to Italian, since translating into morphologically richer languages is a more difficult task [5] compared to the opposite. Italian is a more complex language as it has more verb tenses, and information like gender and number is embedded into words. These elements are not always present into English sentences and must be inferred based on the context.

2.2 Datasets

Two domains were defined to train and evaluate the students.. Dataset *A* [6] contains 175.622 couples of English sentences with their related French translation. These sentences cover everyday life situations, with an average length of 42 characters. The second one, Dataset *B* [7], contains 1.534.699 english sentences with an average length of 133 characters, and are related to educational and academic contexts. Note that

the author of the latter one says that the dataset may introduce some biases, like a more formal writing style. Table 1 shows examples from both datasets.

Source	Example sentences
Dataset A	I'll get something to drink for both of you. I have to do my best. I'm going to wear these shoes on our date tonight.
Dataset B	Muscles, tendons, and ligaments depend upon proper joint movement to function at optimal levels. It is advisable to keep water levels some distance below where the tiles are to prevent any damage. Technological development has also contributed to long-term efficiency and productivity.

Table 1 Comparison of sample sentences from Dataset A and Dataset B.

Then, 35.000 sentences were extracted for training, 5.000 for validation and another 5.000 for testing from each dataset [8], and final datasets were created to be used in the experiments. Since Dataset A consists of English-French pairs, the French side was discarded and only the English sentences were kept.

Next, using OpenAI API, GPT-5 Mini all English sentences were translated into Italian, and the resulting source-target pairs were saved in a CSV file. After completing the translations, an equal number of samples were extracted from Dataset A and Dataset B, and merged into a new Dataset AB, which combined the two domains. GPT-5 Mini was chosen in order to keep costs low; using a more advanced model such as GPT-5.2 would have increased the total cost by approximately 10 times.

Since GPT-5 Mini translations are used as reference labels, the evaluation metrics measure how closely a student has replicated the teacher’s output. A perfect student would have a chrF++ of 100 against the teacher translation; that does not mean that the translation is perfect in an absolute way, but that student can perfectly imitate the teacher.

2.3 Models

The model architectures decided to be used as students come from Google T5-Efficient and mT5 families [9]. These models come with different parameter count, and are pretrained with the Colossal, Cleaned Common Crawl Dataset [10]. The T5 family model and tokenizer are pretrained using only English sentences, while mT5 models are multilingual and support 101 different languages, Italian included.

For this study, both T5 and mT5 variants were tested in order to investigate whether a multilingual model has advantages on cross-lingual knowledge distillation, or if monolingual models with proper finetuning can also achieve similar performance on Italian translation tasks.

In particular, the models used for these experiments are T5-Efficient-Small, T5-Efficient-Base, mT5-Small, and mT5-Base. The parameter count of each model is shown in Table 2.

Name	Parameter
T5-Efficient-Small	60M
T5-Efficient-Base	220M
mT5-Small	300M
mT5-Base	580M

Table 2 Parameter count of the models used.

2.4 Training

Each model listed in Table 2 was fine-tuned on the three different datasets resulting in three variants per base model: Student A, Student B, and Student AB.

In total, 12 models were fine-tuned for 30 epochs on a machine provided by the MIPS Laboratory, equipped with an NVIDIA 3090 (24GB), an Intel i9-12900K, and 64GB of system RAM. All models were trained using 16 bits floating point, batches of size 16, a learning rate of $2e-5$, and chrF++ as metric for evaluating the best model. For a more specific hyperparameter configuration, see code in `train_student.py`. The training of all 12 models took about two days.

3 Results

Figure 1 shows the training and validation loss curves for all 12 models. All fine-tuned models converged within approximately 10 epochs, which was influenced by both model architecture and dataset characteristics. Multilingual models with higher parameter counts (mT5-Small and mT5-Base) converged faster than their monolingual counterparts (T5-Efficient-Small and T5-Efficient-Base). This effect was more pronounced on Dataset B, which contains longer sentences, more technical vocabulary, and greater context-dependency compared to Dataset A, which consists of shorter sentences with simpler vocabulary.

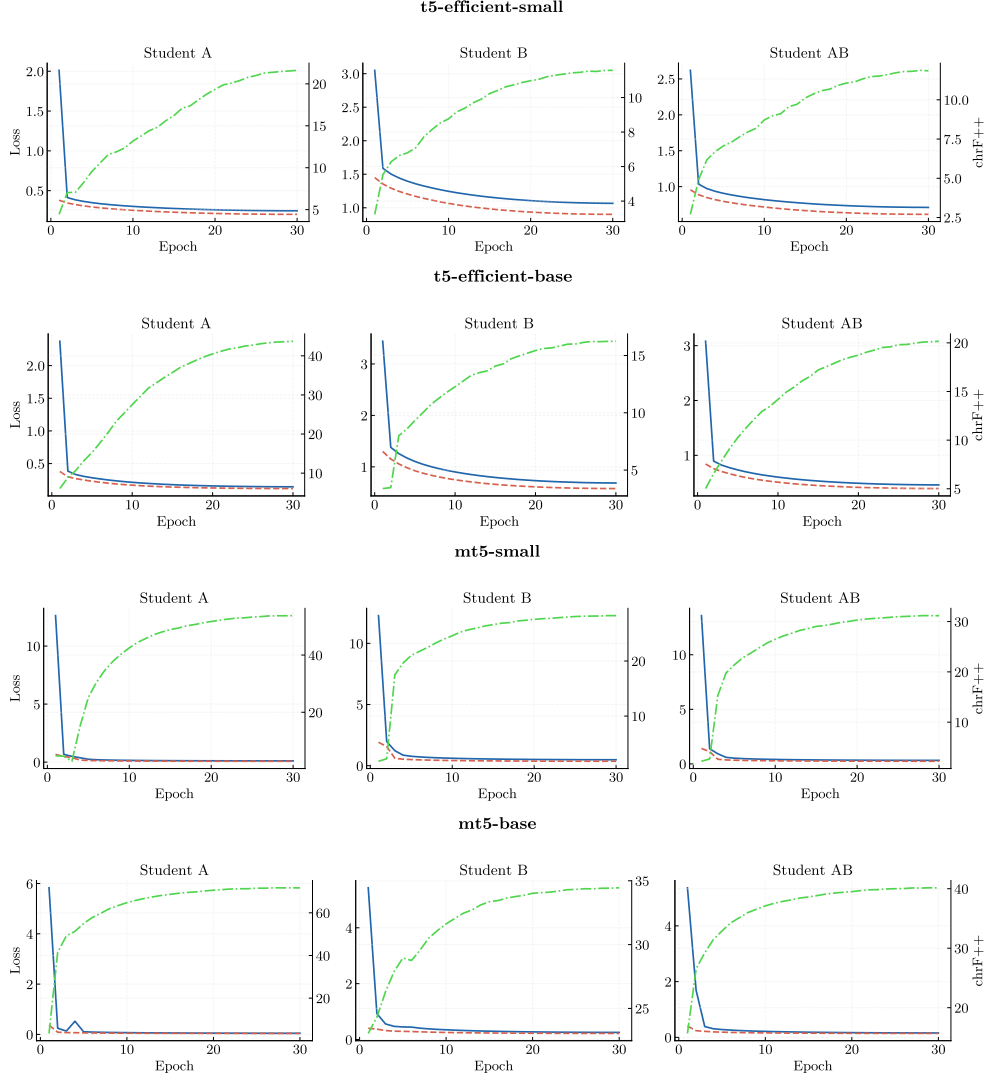


Fig. 1 Training loss (blue line), evaluation loss (orange line), and chrF++ metric (green line) for all models.

3.1 In-domain performance

The chrF++ scores computed on validation data (Figure 1) demonstrate that model size positively correlates with in-domain performance across all datasets. mT5-Base consistently achieves the highest scores, while T5-Efficient-Small shows the lowest performance. This behavior holds for both domain specific models (Student A and Student B) and the mixed domain model (Student AB).

3.2 Out-of-domain generalization

All scores are computed using GPT-5 Mini translations as reference, and measure how well a student can replicate the teacher. In this case the teacher represents the maximum performance achievable, and the gap between 100 and the student scores quantifies the cost of compression.

Table 3 presents both in-domain and out-of-domain performance for all models using chrF++ and BLEU metrics. Bold values indicate the in-domain performance.

Model	Domain	chrF++		BLEU	
		A	B	A	B
T5-Efficient-Small	A	22.1	10.6	4.1	0.4
T5-Efficient-Small	B	17.2	24.6	2	5.2
T5-Efficient-Small	AB	18.2	20.1	2.4	3.3
T5-Efficient-Base	A	45.1	19	23.2	1.7
T5-Efficient-Base	B	26	41.4	6.1	15.4
T5-Efficient-Base	AB	38.1	37.8	16	12.7
mT5-Small	A	53.6	29.8	33.2	6.5
mT5-Small	B	39.7	52.6	16.8	25.9
mT5-Small	AB	49	49.4	27.6	22.4
mT5-Base	A	71.1	45.9	56	19.1
mT5-Base	B	54.3	65.3	32.8	42.5
mT5-Base	AB	66.5	64.2	49.2	40.8

Table 3 In-domain and out-of-domain performance for all models evaluated with chrF++ and BLEU.

Models fine-tuned on a single domain exhibit substantial performance degradation when evaluated on the other domain. For example, mT5-Base trained on Domain A achieves a chrF++ score of 71.1 on Domain A but only 45.9 on Domain B. This domain shift effect is consistent across all model architectures, with smaller models showing more severe degradation. Example translations for one sentence from each dataset are provided in Tables 4 and 5. Table 4 shows translations of the sentence from Dataset A: *"I've heard that some people sleep in bathtubs."*, while Table 5 shows translations of the sentence from Dataset B: *"In 2003, for example, the murder rate in the 12 states without the death penalty was 4.1 for every 100,000 people."*

The Student AB models, trained on the combined dataset, achieve more balanced performance across both domains. While these models do not match the peak in-domain scores of domain-specific models, they demonstrate superior generalization. For instance, mT5-Base trained on Domain AB achieves chrF++ scores of 66.5 and

64.2 on Domains A and B respectively, compared to 71.1 and 45.9 for the domain-specific mT5-Base model trained on Domain A.

Model	Domain	Translation A	Translation B
GPT-5 Mini	-	Ho sentito che alcune persone dormono nelle vasche da bagno.	Nel 2003, ad esempio, il tasso di omicidi nei 12 stati senza pena di morte era di 4,1 ogni 100.000 persone.
T5-Efficient-Small	A	Ho savo che persone sarebbe a scuola in settimana.	In 2003, il tuo è stato a scuola è stato a scuola a scuola è stato a tutti i miei sul tuo giorno a scuola è stato a tutti i giorni a scuola è è stato è stato è stato è stato è stato è stato è stato è stato è stato è stato è stato è stato è stato è stato è stato è stato
T5-Efficient-Small	B	I risultati che alcuni persone saranno in acqua.	In 2003, per esempio, la tua tua tua tua tua tua tua scuola di tuoi 12 paesi che l'infrastruttura era 4.1 per ogni 50,000 persone.
T5-Efficient-Small	AB	Ho che che i persone sul tuo tuo sul tuo tuo sul tuo sul tuo sul tuo sul tuo.	In 2003, per esempio, la rate di murdere nel 12 states non è stato 4.1 per il tuo tuo giorno.
T5-Efficient-Base	A	Ho sentito che qualcuno stanno dormire nel tubo.	In 2003, per esperienza il polizio del 12 polizio è stato risultati a tutti i loro polizio a tutti i loro senza l'incidente delle Stati Uniti è stato 4.1 per tutti i loro polizio delle Stati Uniti che la risulta delle Stati l'è stato il il il il il il anni anni anni anni anni a riso riso riso a tre tre tre anni.
T5-Efficient-Base	B	I hanno spieghi che alcuni persone sai in acqua.	In 2003, per esempio, il tasso di homicide nei 12 stati con il punito di morte era il 4.1 per ogni milioni di persone.
T5-Efficient-Base	AB	Ho sentito che alcuni persone sa senza a scaffè.	Nel 2003, per esempio, la tassa di murdere nei 12 stati con il penalty di morte era 4.1 per ogni milioni di persone.

Table 4 Comparison of example translations from different non-multilingual models for test datasets A and B.

Model	Domain	Translation A	Translation B
GPT-5 Mini	-	Ho sentito che alcune persone dormono nelle vasche da bagno.	Nel 2003, ad esempio, il tasso di omicidi nei 12 stati senza pena di morte era di 4,1 ogni 100.000 persone.
mT5-Small	A	Ho sentito che molte persone dormono in bagni.	In 2003, per quanto l'indice di morte in 12 stati senza la pena di morte è stata 4.1 per ogni 100.000 persone.
mT5-Small	B	Io sono ascoltati che alcuni persone dormono in bagni.	Nel 2003, per esempio, il tasso di morte nei 12 stati senza la pena di morte era di 4.1 per ogni 100.000 persone.
mT5-Small	AB	Ho sentito che alcune persone dormono in bagni.	In 2003, per esempio, il tasso di morte nei 12 stati senza la penalità di morte era 4.1 per ogni 100.000 persone.
mT5-Base	A	Ho sentito che alcune persone dormono nei bagni.	In 2003, per exemple, il tasso di omicidio negli 12 Stati senza la pena di morte era 4.1 per cada 100.000 persone.
mT5-Base	B	Sono sentito che alcune persone dormono nei bagni.	Nel 2003, per esempio, il tasso di omicidio nei 12 stati senza la pena di morte era di 4,1 per ogni 100.000 persone.
mT5-Base	AB	Ho sentito che alcune persone dormono nei bagni.	Nel 2003, per esempio, il tasso di omicidio nei 12 stati senza la pena di morte era di 4,1 per ogni 100.000 persone.

Table 5 Comparison of example translations from different multilingual models for test datasets A and B.

3.3 Monolingual and multilingual models

Despite being pretrained exclusively on English text, T5-Efficient models demonstrate relatively good performances on Italian translation tasks after fine-tuning. T5-Efficient-Base, with 220M parameters, achieves a chrF++ score of 45.1 on Domain A, which is 16% lower than mT5-Small (53.6) despite having 80M fewer parameters and no Italian pretraining. This suggests that task-specific fine-tuning can partially compensate for the lack of multilingual pretraining, even if multilingual models retain a consistent advantage across all experimental conditions.

4 Conclusions

In this work we have discussed the distillation of a student given a black-box teacher in case of translations from English to Italian, and analyzed how domain shift may affect distilled models in two different domains.

These results shows that parameter count is the main indicator of translation quality, with mT5-Base constantly outperforming all other models across all domains. Furthermore, multilingual pretraining gives an advantage over monolingual counterparts, even though some monolingual model have achieved similar performances after finetuning, suggesting that multilingual is beneficial but not a strict necessity when sufficient task specific supervision is available.

We also found out that domain shift is a significant challenge across all models. When a model was trained on a specific domain, performances on the other one dropped. The only solution to this problem is training students with less data but coming from both domains. That mitigates the issue by achieving a balanced performance across all domains, at the cost of an overall modest reduction in scores.

Since student models are trained to reproduce teacher translations, and the evaluation references are themselves generated from GPT-5 Mini, the reported metrics also serve as a direct measure of fidelity to the teacher. The best performing student, mT5-Base trained on Domain AB, achieves chrF++ scores above 64 on both domains, indicating strong knowledge transfer despite the black-box configuration of the teacher.

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